

Prelim Exam Activity

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BSCS - 3A

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Code

```
1  using Avalonia.Controls;
2
3  namespace CS311_Prelim_Exam_Activity_Flores;
4
5  public partial class MainWindow : Window
6  {
7
8      Validators validators;
9      double grossPay = 0.0;
10     double tax = 0.0;
11     double sss = 0.0;
12     double pagibig = 0.0;
13     double philhealth = 0.0;
14     double totalDeductions = 0.0;
15     double netPay = 0.0;
16     double additionalPay = 0.0;
17     double totalPay = 0.0;
18
19
20     public MainWindow()
21     {
22         InitializeComponent();
23         validators = new Validators();
24     }
25
26     private void SubmitButton_Click(object sender,
27     Avalonia.Interactivity.RoutedEventArgs e)
28     {
29         if (validators.validateTextBox(in RateTextBox) is string s)
30         {
31             DataValidationErrors.SetError(RateTextBox, new CleanException(s));
32         }
33         else
34         {
35             DataValidationErrors.ClearErrors(RateTextBox);
36         }
37         if (validators.validateTextBox(in HoursTextBox) is string s2)
38         {
```

```

39         DataValidationErrors.SetError(HoursTextBox, new
           CleanException(s2));
40     }
41     else
42     {
43         DataValidationErrors.ClearErrors(HoursTextBox);
44     }
45
46     if (validators.countErrors() == 0)
47     {
48         double rate = double.Parse(RateTextBox.Text!);
49         double hours = double.Parse(HoursTextBox.Text!);
50
51         grossPay = rate * hours;
52         tax = grossPay * 0.12;
53
54         if (cbSSS.IsChecked! == true) sss = grossPay * 0.10; else sss = 0;
55         if (cbPagibig.IsChecked! == true) pagibig = grossPay * 0.08; else
           pagibig = 0;
56         if (cbPhilhealth.IsChecked! == true) philhealth = grossPay * 0.06;
           else philhealth = 0;
57
58         totalDeductions = sss + pagibig + philhealth;
59         netPay = grossPay - totalDeductions;
60
61         if (rbPermanent.IsChecked! == true) additionalPay = 500; else
           additionalPay = 0;
62
63         totalPay = netPay + additionalPay;
64
65         GrossPayTextBox.Text = $"{grossPay:.2}";
66         TaxTextBox.Text = $"{tax:.2}";
67
68         SSSTextBox.Text = sss == 0.0 ? "" : $"{sss:.2}";
69         PagibigTextBox.Text = pagibig == 0.0 ? "" : $"{pagibig:.2}";
70         PhilhealthTextBox.Text = philhealth == 0.0 ? "" :
           $"{philhealth:.2}";
71
72         TotalDeductionsTextBox.Text = $"{totalDeductions:.2}";
73         NetPayTextBox.Text = $"{netPay:.2}";
74         AdditionalPayTextBox.Text = $"{additionalPay:.2}";
75         TotalPayTextBox.Text = $"{totalPay:.2}";
76     }
77 }
78
79 private void ClearButton_Click(object sender,
   Avalonia.Interactivity.RoutedEventArgs e)
80 {

```

```

81         RateTextBox.Text = "";
82         HoursTextBox.Text = "";
83
84         GrossPayTextBox.Text = "";
85         TaxTextBox.Text = "";
86
87         SSSTextBox.Text = "";
88         PagibigTextBox.Text = "";
89         PhilhealthTextBox.Text = "";
90
91         TotalDeductionsTextBox.Text = "";
92         NetPayTextBox.Text = "";
93         AdditionalPayTextBox.Text = "";
94         TotalPayTextBox.Text = "";
95
96
97         grossPay = 0.0;
98         tax = 0.0;
99         sss = 0.0;
100        pagibig = 0.0;
101        philhealth = 0.0;
102        totalDeductions = 0.0;
103        netPay = 0.0;
104        additionalPay = 0.0;
105        totalPay = 0.0;
106        DataValidationErrors.ClearErrors(RateTextBox);
107        DataValidationErrors.ClearErrors(HoursTextBox);
108    }
109 }

```

Video

https://drive.google.com/file/d/1Jc-CcFJevc-LhK47GU_NKpSy7lsNFQ5B/view?usp=sharing